

SAINT-YAN AP, France

WMO: 073790

Lat: 46.4083N

Lon: 4.0164E

Elev: 243

StdP: 98.44

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.0	(c) -4.8	(d) -10.0	(e) 1.7	(f) -5.2	(g) -7.2	(h) 2.1	(i) -3.2	(j) 10.3	(k) 10.1	(l) 9.0	(m) 9.4	(n) 1.4	(o) 60	(p) 0.487

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.6	33.0	20.5	30.6	20.0	28.5	19.3	21.7	29.5	20.9	28.5	20.1	27.0	2.8	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.4	14.5	24.6	18.5	13.8	23.4	17.8	13.1	22.4	64.4	29.7	61.4	28.7	58.5	27.1	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.7	(b) 7.3	(c) 6.3	(e) -11.1	(f) 35.7	(g) 5.9	(h) 2.1	(i) -15.4	(j) 37.2	(k) -18.8	(l) 38.4	(m) -22.1	(n) 39.6	(o) -26.4	(p) 41.1		
(a) 8.7	(b) 7.3	(c) 6.3	(e) -11.3	(f) 23.6	(g) 5.6	(h) 1.1	(i) -15.3	(j) 24.4	(k) -18.6	(l) 25.0	(m) -21.7	(n) 25.6	(o) -25.8	(p) 26.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.4	3.5	4.2	7.4	10.3	14.2	18.0	20.0	19.6	15.6	12.2	7.0	4.0
	DBStd	6.96	4.33	4.08	3.58	3.34	3.39	3.68	3.28	3.38	3.22	3.85	3.94	4.36
	HDD10.0	822	203	164	95	36	5	0	0	0	1	22	106	189
	HDD18.3	2732	459	396	338	241	136	51	19	24	92	191	341	444
	CDD10.0	1323	2	2	14	46	134	239	309	297	171	91	14	4
	CDD18.3	193	0	0	0	0	6	40	71	63	12	2	0	0
	CDH23.3	2416	0	0	0	15	114	521	831	755	163	17	0	0
	CDH26.7	899	0	0	0	1	22	188	328	321	38	1	0	0
Wind	WSAvg	2.3	2.5	2.8	2.7	2.6	2.3	2.2	2.2	2.0	1.8	2.0	2.4	2.5
Precipitation	PrecAvg	877	62	54	52	69	90	79	78	78	77	82	90	66
	PrecMax	1059	125	102	145	157	188	197	165	147	156	172	191	133
	PrecMin	659	24	11	11	19	20	29	22	20	23	16	27	22
	PrecStd	112	27	26	29	36	39	40	35	34	37	33	36	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.5	16.6	21.2	25.3	29.2	34.8	35.6	36.8	30.4	25.6	19.1	15.7
		MCWB	10.9	10.7	13.3	16.0	18.9	21.2	20.3	20.7	18.6	17.6	14.3	11.4
	2%	DB	12.8	14.0	18.8	22.4	26.5	31.2	32.6	32.9	27.4	22.9	16.8	13.6
		MCWB	10.0	9.6	11.8	14.2	17.8	20.4	20.6	20.4	18.4	16.8	13.0	10.7
	5%	DB	11.4	12.2	16.4	20.3	24.1	28.6	30.3	30.2	25.0	20.9	14.9	12.2
		MCWB	8.9	8.6	10.8	13.3	16.6	19.6	20.1	20.0	17.8	15.9	12.2	10.0
	10%	DB	9.9	10.6	14.3	17.8	22.0	26.2	28.1	27.6	22.8	18.9	13.2	10.6
		MCWB	8.0	8.0	9.8	12.0	15.7	18.7	19.2	19.2	16.8	15.0	11.0	8.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.7	11.4	14.1	17.0	20.3	23.0	22.8	22.8	20.9	18.8	14.9	12.2
		MCDB	13.8	14.9	19.9	23.4	27.0	31.3	30.4	30.8	26.6	23.5	17.7	14.2
	2%	WB	10.3	10.1	12.4	15.2	18.8	21.6	21.6	21.5	19.4	17.4	13.5	11.1
		MCDB	12.3	13.0	17.0	21.6	24.7	28.9	29.6	29.7	25.3	21.6	15.9	13.0
	5%	WB	9.1	9.1	11.2	13.7	17.4	20.4	20.7	20.6	18.3	16.4	12.4	10.1
		MCDB	11.1	11.7	15.4	18.8	22.6	27.0	28.3	28.5	23.7	20.1	14.6	12.0
	10%	WB	8.0	8.1	10.2	12.5	16.2	19.3	19.8	19.7	17.3	15.3	11.2	8.9
		MCDB	9.7	10.3	13.8	16.8	20.9	24.9	26.8	26.4	21.7	18.2	12.9	10.5
Mean Daily Temperature Range	5% DB	MDBR	6.8	8.2	10.8	11.9	11.8	12.5	12.6	12.9	12.0	10.3	7.7	6.7
		MCDBR	8.8	11.9	15.4	16.8	16.3	17.2	17.2	18.3	16.3	13.4	10.2	8.6
		MCWBR	6.4	8.1	9.4	9.5	8.5	7.5	6.6	6.8	8.0	7.8	6.9	6.2
	5% WB	MCDBR	8.1	10.6	13.5	15.3	14.4	15.0	15.2	15.5	14.0	12.3	9.7	8.2
		MCWBR	6.2	7.5	8.5	8.9	8.0	7.1	6.4	6.6	7.2	7.7	7.0	6.2
Clear-Sky Solar Irradiance	taub	0.310	0.330	0.358	0.393	0.397	0.405	0.400	0.395	0.369	0.358	0.325	0.306	
	taud	2.461	2.403	2.337	2.244	2.284	2.300	2.308	2.335	2.382	2.413	2.459	2.463	
	Ebn at Noon	790	841	865	862	869	862	860	851	842	795	760	750	
	Edh at Noon	68	87	107	129	128	127	124	116	102	85	68	61	
All-Sky Solar Radiation	RadAvg	1.15	1.98	3.25	4.44	5.20	6.10	5.89	5.13	3.96	2.46	1.34	0.96	
	RadStd	0.14	0.30	0.41	0.58	0.51	0.55	0.48	0.42	0.29	0.20	0.14	0.15	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (28 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page